Reg. No. | | | | | | | | | | | [14EE61]

SIXTH SEMESTER B. Tech EXAMINATION, JUNE–2019

POWER SYSTEM ANALYSIS – I

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions choosing atleast TWO full question from each part

PART-A

1. a. Define per unit computation and mentions its advantages. **04**

b. Draw the reactance diagram of the system shown in fig below. The ratings of the components are **10**

G : 15MVA, 6.6kV, X" = 12%

T1 = 20 MVA, 6.6/66 kV, X=8%

T2 = 20 MVA, 66/6.6 kV, X=8%

M1 & M2 : 5MVA, 6.6kV, X"=20%

Consider the ratings of the generator as base values.

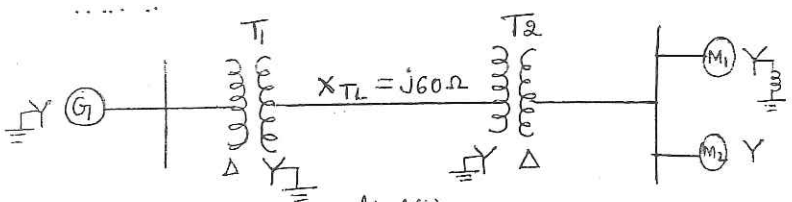

fig 1(b)

2. a. Describe the transients in a transmission line on the occurrence of a short circuit. **08**

b. A synchronous generator and motor are rated 30,000 KVA, 13.2 kV and both have sub transient reactance of 20%. The line connecting them has a reactance of 10% on the base of machine ratings. The motor is drawing 20,000 KW at 0.8 p.f leading. The terminal voltage of the motor is 12.8 kV. When a symmetrical 3 phase fault occurs at motor terminals, find the sub transient current in generator, motor and at fault point. **06**

3. a. Obtain the expression for Symmetrical components in terms of phase voltages **07**

b. In a 3 phase system, the positive, negative and zero sequence components of line currents are $I_{a1} = 100\angle 30^\circ$ A, $I_{b2} = 40\angle 90^\circ$ A and $I_{c0} = 10\angle -30^\circ$ A. Find the line currents. **07**

4. a. Draw positive , negative and zero sequence networks for the power system shown **10**

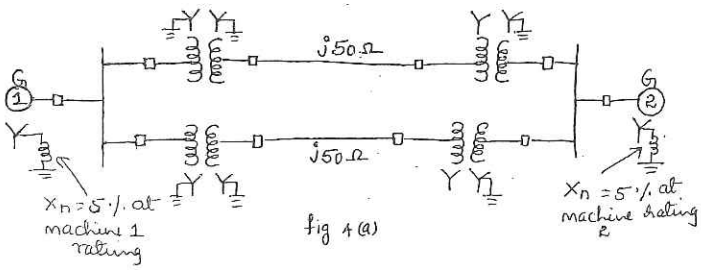

fig 4(a)

Choose a base of 50 MVA, 220kV in the 50Ω transmission lines and mark all reactances in p.u. The ratings of the generators and transformers are:

G_1 : 25MVA, 11kV, X" = 20%

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G_2 : 25 MVA, 11kV , X'' =20%

Three phase transformer (each) : 20MVA, 11 Y/ 220 Y kV , X= 15%

The negative sequence reactance of each synchronous machine is equal to the sub transient reactance. The zero sequence reactance of each machine is 8%. Assume that the zero sequence reactances of lines are 250% of their positive sequence reactances.

b. Draw zero sequence networks for the following 3 ϕ transformers. **04**

i. Y- Δ **ii.** Y - Δ **iii.** Y - Y **iv)** Δ - Δ

PART-B

5. a. Derive an expression for fault current, when double line-to-ground fault occurs on an unloaded generator. **07**

b. A three phase generator with an open circuit voltage of 400V is subjected to an LG fault through a fault impedance of j2Ω. Determine the fault current if Z_1= j4 Ω, Z_2 = j2Ω, Z_0 = j1 Ω. Repeat the problem for L – L fault having same fault impedance. **07**

6. a. A synchronous motor is receiving 10MW of power at 0.8 p.f. lag at 6KV. An LG fault takes place at the middle of the transmission line as shown in fig below. Find the fault current. The ratings of the generator, motor and transformer are as follows: **14**

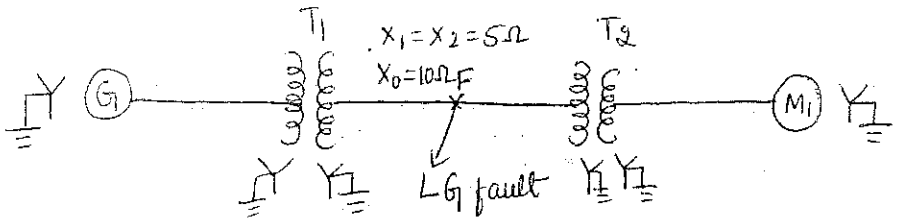
Generator: 20MVA, 11kV, X1 = 0.2p.u, X_2 = 0.1 p.u, X_0 = 0.1 p.u.

Transformer T1: 18MVA, 11.5 kV Y – 34.5kV Y, X = 0.1 p.u.

Transmission line: X_1= X_2 = 5Ω, X_0 = 10Ω.

Transformer T2: 15MVA, 6.9kV Y – 34.5kV Y, X = 0.1 p.u.

Motor: 15 MVA, 6.9kV, X_1 = 0.2 p.u., X_2=X_0 = 0.1 p.u.



7. a. Derive the power angle equation of a non-salient pole alternator connected to infinite bus. Also draw the power angle curve. **07**

b. Analyze swing equation corresponding to rotor dynamics in a synchronous machine. **07**

8. a. Derive equal area criterion for a machine connected to infinite bus bar and Mention the applications of equal area criterion. **08**

b. An equivalent generator connected to 50 Hz infinite bus has steady state power limits before, during and after a fault is cleared as 2.0 pu, 0.5 pu and 1.5 pu. Calculate the critical clearing angle if the initial load is 1.0 pu. **06**